Q.19 What is the benefit of frequency translation in communication system? (CO1)

Q.20 Define sampling theorem and explain its significance. (CO4)

Q.21 With the help of block diagram, explain FSK system. (CO5)

Q.22 What is the effect of noise on FM carrier? (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Derive the expression for an amplitude-modulated (AM) wave and explain the significance of carrier and sidebands. Also draw the waveforms. (CO2)

Q.24 Explain basic block diagram of QPSK and its working principle. (CO5)

Q.25 Draw and explain the block diagram of a basic digital communication system. (CO4)

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3rd Sem / ECE, ECE (For Speech and Hearing Impaired)

Subject : Analog and Digital Communication

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 VHF range is ________. (CO1)

a) 3- 30 MHz b) 300-3000 MHz

c) 30-300 MHz d) 3- 30 KHz

Q.2 The modulation index of an AM wave is defined as: (CO2)

a) Ratio of carrier frequency to modulating frequency

b) Ratio of modulating signal amplitude to carrier amplitude

c) Ratio of carrier power to sideband power

d) None of the above

Q.3 In frequency modulation, the modulation index is inversely proportional to: (CO3)

a) Carrier amplitude

b) Modulating signal amplitude

c) Carrier frequency

d) Modulating signal frequency

Q.4 The maximum bandwidth is occupied by (CO5)

a) ASK b) BPSK

c) FSK d) None of the above

Q.5 Give full form of DPCM (CO4)

a) Discrete Pulse Code Modulation

b) Digital Pulse Code Modulation

c) Differential Pulse Code Modulation

d) Dynamic Pulse Code Modulation

Q.6 The process of signal compression and expansion is called _____. (CO4)

a) Amplification b) Modulation

c) Companding d) Compression

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The bandwidth of an AM signal is______. (CO2)

Q.8 In FM, the noise triangle represents_____. (CO3)

Q.9 In sampling theorem the relationship between sampling frequency and modulating frequency is given by the formula ______. (CO4)

Q.10 Give full form of PSK modulation_____. (CO5)

Q.11 The primary advantages of using FM over AM is______. (CO3)

Q.12 In digital communication, quantization introduces ______. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the need for modulation in communication systems. (CO1)

Q.14 Explain working principle of PPM. (CO4)

Q.15 What is noise? What are different types of noise observed in communication system? (CO4)

Q.16 What is the difference between DSB-SC and SSB-SC modulation? (CO2)

Q.17 Explain the role of limiter in FM receivers. (CO3)

Q.18 Compare FM and AM modulation techniques. (CO3)